

Analytical Workflows

IB 516

Developed by Mark Novak and Ben Dalziel

Dept. of Integrative Biology
Oregon State University

Overview

Analytical
Workflows

Philosophy

Course
Outline

Website & Schedule

Syllabus

Accountabilibuddies

Accounts &
Apps

Project
Proposals

Our hope

1 Philosophy

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Automation

Doing science involves repetition at multiple levels.
Automation is more efficient and reduces mistakes.

Reproducibility

Science must be reproducible, by others and you.
Your work should be *transparent, organized, and self-contained*.

Correctness

Science is done by humans; mistakes are inevitable.
We must work in ways that protect against bugs.

¹See Allesina & Wilmes 2019

Not required reading

But good! Currently missing from the Valley library...

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Computing Skills for Biologists

A TOOLBOX



Stefano Allesina and Madlen Wilmes

Further optional reading

Helpful for seeing how pipelines are built step by step. Also really fun. Lots online: natureofcode.com

THE NATURE OF CODE BY DANIEL SHIFFMAN

♥ SUPPORT

🔗 GITHUB

🧑 CODING TRAIN

Order Direct

Dedication

Acknowledgments

Introduction

0 Randomness

1 Vectors

2 Forces

3 Oscillation

4 Particle Systems

5 Autonomous Agents

6 Physics Libraries

7 Cellular Automata



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Simplicity

Analyses should be coded as simply as possible.
The clearest solution is often the best one.

Openness

Science is a worldwide endeavor.
Access to it should be free and equitable.

Science as Software Development

Science is a *process* of versions and updates.
Software developers have tools to help.

¹See Allesina & Wilmes 2019

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`github.com/analyticalworkflows/TeachingMaterials`

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- Weekly check-in
 - *goals/deadlines, accomplishments, sticking-points*
 - *final 10 min. of every 2nd class*
- Randomized pairing
 - *re-randomized each week*

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What you'll need

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Accounts

- GitHub¹

Software

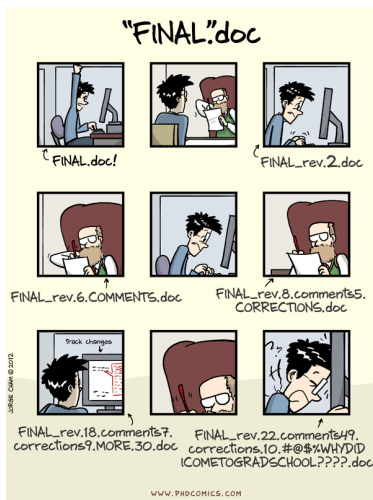
- An integrated development environment (IDE), such as Visual Studio Code (VS Code), RStudio, etc.

Git interface

- GitHub Desktop, VS Code, RStudio, Sourcetree, etc.

L^AT_EX editor

- Stand-alone: TeXstudio, VS Code, etc.
- Online: Overleaf, Papeeria, TeXpage, etc.



¹Please email me your username.

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Primary Goal

To develop more effective research skills.

Secondary Goal

To make significant progress on your thesis.

Our belief

We can achieve both in this course!

Let's get started!

.../TeachingMaterials/classes/ProjectProposal

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We're all learning

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- Use this course as a workshop for your thesis
- Lean into the work—you and your peers will benefit
- Be kind to yourself
- Support each other